11-26 第2篇

阅读理解（说明文） Why Do Wild Geese Fly in a V-Shape?

① Every autumn, just before the first serious cold, you can sometimes see, in the grey morning sky above our city, a thin **V-shape** of dark dots moving slowly south. If you are quiet, you will also hear a low, almost lonely **"honk, honk"** floating down. Those are **wild geese**, travelling from the cold north toward warmer lakes and rice fields, thousands of kilometres away. A small puzzle has followed these birds for centuries. Why do they always fly in that careful V-shape, and not in a straight line, a circle, or a random cloud? The answer is surprisingly precise: the V-shape is a piece of **living physics**, and the birds have been using it for longer than we have been watching them.

② The first reason lies in **what the wings do to the air**. When a goose beats its wings, it pushes air downward under each wing to stay up. But, as a small side-effect of this, a **small rising current of air** — called an **upwash** — forms just off the tip of each wing. If a second goose places itself at exactly the right spot, a little **behind and to the side** of the bird in front, its wings can catch the upwash and save energy. In the V-formation, **each goose (except the leader)** flies inside the soft rising air that the goose in front has kindly made. It is a little like drafting in a cycling team, or like running in the quiet shelter behind a tall friend in strong wind.

③ How much energy does this save? In 2001, a French scientist, **Henri Weimerskirch**, used small heart-rate monitors on pelicans flying in V-shape and found that each bird's heart beat **11% to 14% slower** when flying in formation than when flying alone. Later studies on geese, estimated from wing-beat counts and wind-tunnel models, suggest a similar or even larger saving — up to **20%—30%** extra flight time on the same amount of food. Over a 3,000-kilometre migration, that small percentage means **a hundred kilometres of free flight** per goose. For a tired bird on a cold October morning, it is not a luxury; it is the difference between reaching the lake and falling short.

④ The second reason is about **seeing each other**. A goose's eyes are on the side of its head, giving it a very wide view. When the flock is in a V, each bird can see **both the leader and every goose beside it**, without being blocked by a wall of wings. This is why you rarely see geese flying in a perfect straight line; a straight line, from the back, hides everyone in front. A V, in contrast, gives **every member a clear view**, helps the flock to keep spacing, and allows each bird to notice quickly when another is in trouble — slowing down, climbing, or drifting off course.

⑤ The third reason is **communication**. A V-shape also makes calling efficient. The famous "honk" of geese is not just loud talking; studies from 2014 at the University of

Exeter suggest that the low honks travel better **along the line of the V** and serve to encourage the leader — a kind of friendly "keep going!" shout, repeated every few seconds by the birds behind. A random cloud of geese cannot easily pass this signal forward and back; a tidy V can.

⑥ A small surprise: the leader of the V does **not** stay the leader for the whole day. Because the front bird has no upwash to catch and does the hardest work, leaders rotate regularly. A 2015 study by Steven Portugal and colleagues placed tiny loggers on Northern bald ibises (another V-flying bird) and found that **the lead position was shared carefully** — each bird taking a turn at the front for a few minutes, then dropping back to rest in the soft upwash. Among geese, the same "**fair turn-taking**" seems to happen. There is no single heroic captain leading from sunrise to sunset; there is a **rotating team**, each member quietly paying its share of the wind.

⑦ Another small fact: **sick or weak birds** do not get left behind. When one goose cannot keep up, **two or three other geese** will sometimes leave the main V and drop down with it, flying in a small separate formation until either the weak goose recovers or — sadly — cannot continue. Then the small group may try to catch up with the main flock at the next stop. This was documented by bird-watchers in Alberta, Canada, in 2019. It is not a rule, and it is not always seen; but when it does happen, it is one of the most quietly moving sights in nature.

⑧ A few practical details: the angle of the V is usually about **45 degrees**, though it changes with wind. In very strong wind, the V flattens; in very calm air, it sharpens. The birds do not, of course, calculate the angle; their wings **feel** the exact sweet spot. Over thousands of years, geese with a weaker feel for this position simply flew home more tired and fed less well, and, over generations, left fewer children. The careful V we see today is the polished result of **millions of small choices about a small amount of lift** — repeated by every migrating goose, every autumn, for a very long time.

⑨ So the next time you look up on a cold November morning and see a thin V moving south across a pale sky, know that you are seeing three things at once: a **formula of physics** in feathers, a **quiet team** sharing the hardest position, and a **long memory** of the wind passed down, bird to bird, for longer than any human library has existed. The honks above you are not just lonely sounds. They are the rolling thank-yous of a small society that long ago solved a cold problem by flying, very politely, at an angle.

注释： upwash 上升气流；drafting 利用前者挡风；migration 迁徙；heart-rate 心率；ibis 朱鹮

根据短文内容，选择最佳答案。

1. According to Paragraph ②, how does flying in a V-shape save energy?

A. By making the birds heavier.
B. When a goose beats its wings, a small rising current of air called an "upwash" forms just off each wing tip; geese behind and slightly to the side of the leader can catch this upwash and save energy — similar to drafting in a cycling team.
C. By warming the air with their wings.
D. By moving in circles.

2. How much energy can flying in V-formation save, according to research?

A. Only about 1%.
B. About 50% always.
C. French scientist Henri Weimerskirch's 2001 pelican study found each bird's heart beat 11%—14% slower in formation; later studies on geese suggest 20%—30% extra flight time on the same amount of food — about 100 km of "free flight" over a 3,000 km migration.
D. There is no saving at all.

3. Why do geese rarely fly in a perfectly straight line, according to Paragraph ④?

A. Because a straight line is illegal.
B. Because a goose's eyes are on the side of its head; in a straight line, all birds in front are hidden from those behind — a V-shape gives every member a clear view of the leader and every neighbour, helping keep spacing and spot any bird in trouble.
C. Because a straight line is too easy.
D. Because they cannot see at all.

4. What does the 2015 study of Northern bald ibises by Steven Portugal show about leadership in V-flight?

A. A single heroic captain leads from sunrise to sunset.
B. The lead position is carefully shared — each bird takes a turn at the front for a few minutes and then drops back to rest in the soft upwash; among geese, the same fair turn-taking seems to happen.
C. Only the strongest male leads.
D. The leader is chosen by voting.

5. What does the writer mean by calling the V-shape "a formula of physics in feathers, a quiet team, and a long memory"?

A. That physics and biology have nothing to do with each other.
B. That, in one V-shape, you are seeing at once: a careful physics of lift and air, a quiet team sharing the hardest leading position, and a long memory of how to fly well, polished over millions of years and passed from bird to bird.
C. That V-shapes are a coincidence.
D. That geese invented mathematics.

答案与解析

1.
【答案】B
【题型】细节理解题
【定位】第②段 upwash 原理
【解析】直引 + 自行车跟骑类比。A、C、D 均错。故选 B。

2.
【答案】C
【题型】细节理解题
【定位】第③段 Weimerskirch 与数据
【解析】直引。A、B、D 均错。故选 C。

3.
【答案】B
【题型】细节理解题
【定位】第④段视野分析
【解析】侧眼 + 直线遮挡直引。A、C、D 均错。故选 B。

4.
【答案】B
【题型】细节理解题
【定位】第⑥段 Portugal 研究
【解析】"公平轮换"直引。A、C、D 均错。故选 B。

5.
【答案】B
【题型】推理判断题
【定位】第⑨段结尾
【解析】"物理 + 团队 + 记忆"三重寓意主旨。A、C、D 均偏离。故选 B。